Aiden Cary Phase 4 Discussion

After gathering the data, I noticed that both the Calinski-Harabasz (CH) and Silhouette Width (SW) indices were unable to estimate the true number of clusters (according to the true K values given in Phase 2) for every dataset. However, this is expected because these indices are estimates based on the partition that is generated by the best run of K-means (the run with the lowest SSE), since they are both maximization indices.

Dataset	True K	Best K (CH)	Best K (SW)
Ecoli	8	4	4
Glass	6	2	3
Ionosphere	2	2	4
Iris Bezdek	3	3	2
Landsat	6	3	3
Letter Recognition	26	2	85
Segmentation	7	4	2
Vehicle	4	2	2
Wine	3	2	3
Yeast	10	2	4

Looking at each dataset individually:

- ecoli dataset (true K=8):
    - Both CH and SW estimated K=4. Neither index detected the true number of clusters.
    - The CH score peaked at K=4 (204.8932) and decreased steadily afterward, while SW also peaked at K=4 (0.4270).
- glass dataset (true K=6):
    - CH estimated K=2 and SW estimated K=3. Neither was correct, but SW was closer to the true value.
    - The CH score dropped sharply from K=2 (144.2997) to K=3 (106.215) and continued to drop steadily afterward.
    - SW was nearly identical at K=2 (0.5241) and K=3 (0.5252), with K=3 edging out K=2 only by 0.0011.
- ionosphere (true K=2):
    - CH correctly estimated K=2, while SW estimated K=4 incorrectly.
    - The CH score was highest at K=2 (115.036) and decreased at every subsequent K. SW continued rising to a peak at K=4 (0.3047) from its K=2 value (0.2934), suggesting it found a better partition by splitting one of the two clusters further.
- iris_bezdek (true K=3):
    - CH correctly estimated K=3 while SW estimated K=2 incorrectly.

    - The CH scores at K=2 (354.3656) and K=3 (359.8451) were very close, with K=3 narrowly winning. SW dropped sharply from K=2 (0.6300) to K=3 (0.5048), causing it to prefer the 2-cluster partition.
- landsat (true K=6):
    - Both CH and SW estimated K=3. Neither index detected the true number of clusters.
    - CH peaked at K=3 (5844.3918) and fell at K=4 (5323.097), and SW similarly peaked at K=3 (0.4321) and fell at K=4 (0.4028).
- letter_recognition dataset (true K=26):
    - CH estimated K=2 and SW estimated K=85. Both indices were far off from the true K in two opposite directions, but CH was closer to the true value.
    - The CH score was highest at K=2 (4748.0391) and decreased steadily for every K tested.
    - SW scores were uniformly low and crept upward through K=85 (0.1824) without ever finding a clear peak.
- segmentation (true K=7):
    - CH estimated K=4, and SW estimated K=2. Neither index detected the true number of clusters, but CH was closer to the true value.
    - CH found a local maximum at K=4 (1843.7227), which was higher than both K=2 (1783.559) and K=3 (1670.5321).
    - SW peaked at K=2 (0.3892) and declined steadily afterward.
- vehicle (true K=4):
    - Both CH and SW estimated K=2. Neither index detected the true number of clusters.
    - The CH score at K=2 (702.5785) dropped sharply to K=3 (549.866) and continued falling.
    - SW similarly peaked at K=2 (0.4033) and dropped to 0.3040 at K=3.
- wine (true K=3):
    - CH estimated K=2 while SW correctly estimated K=3.
    - The CH scores at K=2 (84.7085) and K=3 (83.3737) were nearly identical, with K=2 winning by only 1.3348.
    - SW also showed very similar values at K=2 (0.2987) and K=3 (0.3013), with K=3 edging out K=2 by only 0.0026.
- yeast (true K=10):
    - CH estimated K=2 and SW estimated K=4. Neither was correct, but SW was closer to the true value.
    - The CH score was highest at K=2 (386.46) and fell steadily, while SW peaked at K=4 (0.2499).

Overall, the Calinski-Harabasz index performed better across these datasets. CH correctly identified the true K in 2 out of 10 datasets (ionosphere and iris_bezdek), while SW only correctly identified 1 (wine). For the datasets where neither index was correct, CH produced a closer estimate in 2 of the remaining 7 (letter_recognition and segmentation), compared to SW's 2 (glass and yeast), with the other 3 datasets being ties. CH was also easier to interpret by its tendency to produce a more decisive and noticeable peak. SW was more prone to peaking too early at K=2 or, for the letter_recognition dataset, continuing to rise without ever finding a clear peak.